Evaluation Report – Recommender System

1. Introduction

This report evaluates the performance of a recommender system built using user interaction data. The system generates product recommendations using multiple approaches:

- Popularity-Based Recommendation
- Collaborative Filtering (User-Based)
- Matrix Factorization using SVD

The goal of this evaluation is to compare these models and understand their effectiveness in generating relevant recommendations.

2. Methodology

2.1 Data Preparation

The dataset consists of processed user interaction data from Week 1:

- products_clean.parquet
- user_behavior.parquet

User interactions were converted into numerical scores:

- View → 1
- Cart → 5
- Purchase → 10

Only active users (with more than 5 interactions) were considered to reduce noise.

2.2 Train-Test Split

The dataset was split as follows:

- Training Set: 80%
- Test Set: 20%

Example:

- Train size: 23982 records
- Test size: 5996 records

2.3 Evaluation Metrics

The following metrics were used:

- **Precision@10**: Measures how many recommended items are relevant
- **Recall@10**: Measures how many relevant items are successfully recommended
- **MAP@10 (Mean Average Precision)**: Evaluates ranking quality

3. Results

Model	Precision@10	Recall@10	MAP@10
Popularity	0.00643	0.01873	0.00606
Collaborative Filtering	0.00627	0.01974	0.00495
SVD Model	0.00606	0.02095	0.00670

4. Observations

- The **Popularity Model** achieves the highest precision, indicating that frequently interacted products are more likely to be relevant.
- The **SVD Model** performs best in terms of Recall and MAP, showing its ability to capture latent relationships between users and products.
- The **Collaborative Filtering model** performs moderately but is limited by sparse user interaction data.
- All models show relatively low metric values due to the nature of implicit feedback data, where user preferences are not explicitly defined.
- The results demonstrate a trade-off between precision and recall across different models.

5. Sample Recommendations (Optional Insight)

Example outputs from the system:

Collaborative Filtering

- Doctor Bed
- Compare Bed
- Election Wardrobe
- Occur Table

SVD-Based Recommendations

- Doctor Bed
- Family Bed
- World Chair
- Option Wardrobe

Final Output (Popularity-Based)

- Detail Wardrobe
- Magazine Sofa

- Enjoy Table
- Nature Wardrobe

These examples illustrate how different models recommend different types of products based on their logic.

6. Conclusion

The evaluation shows that:

- The **SVD-based model** provides the best overall recommendation quality due to improved ranking and coverage.
- The **Popularity model** remains competitive in precision due to strong global trends.
- The **Collaborative Filtering model** is limited by sparse data and weaker user similarity signals.

Despite relatively low metric values, the system successfully demonstrates a complete recommendation pipeline and highlights real-world challenges such as data sparsity and implicit feedback limitations.

7. Future Improvements

- Introduce time-based data splitting for more realistic evaluation
- Implement hybrid recommendation models (content + collaborative)
- Improve feature engineering for stronger user signals
- Deploy the system as a real-time recommendation API
